CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Effective communication within educational institutions is essential for the smooth functioning of academic and administrative activities. Universities and colleges regularly issue circulars to inform students and staff about important announcements such as examination schedules, academic events, placement drives, holidays, and departmental updates. Traditionally, these circulars are distributed through notice boards, printed documents, emails, or messaging groups. However, these manual or semi-digital methods often lead to issues such as delayed communication, lack of centralized access, misplacement of information, and difficulty in tracking whether recipients have received or acknowledged the circulars.

The rapid advancement of cloud technologies and digital platforms, organizations are increasingly adopting modern solutions to streamline internal communication and information management. Cloud-based systems provide centralized data storage, real-time access, and improved security, making them highly suitable for managing institutional notifications and announcements. By leveraging such technologies, educational institutions can replace traditional circular distribution methods with automated, efficient, and transparent systems that ensure timely delivery of information to the intended recipients.

The Department Circular Management System is developed as a cloud-based solution using the Salesforce Platform, which provides a scalable and secure environment for managing circular creation, approval, and distribution. The system enables administrators to create and manage circulars digitally, while incorporating an approval workflow where department heads review and approve the circulars before they are published. Once approved, the circulars are automatically made available to the appropriate audience such as students or staff, ensuring that the right information reaches the right users at the right time.

The system integrates Salesforce features such as Experience Cloud portals, automated email notifications, role-based access control, and approval processes to enhance communication efficiency. Students and staff can access circulars through a centralized digital portal, view important announcements relevant to them, and acknowledge the circulars if required. By digitalizing the circular management process and integrating it with modern cloud technologies, the Department Circular Management System improves transparency, reduces manual effort, and ensures reliable communication across the institution. This project demonstrates how cloud-based CRM platforms like Salesforce can be effectively utilized to modernize institutional communication systems and create a more organized and efficient academic environment.

1.2 EXISTING SYSTEM

The existing circular management system in many educational institutions primarily operates through traditional and semi-manual communication methods. Important announcements such as examination schedules, academic events, departmental meetings, placement drives, and holiday notices are usually shared through physical notice boards, printed circulars, emails, or messaging platforms. Administrative staff or faculty members prepare these circulars manually and distribute them through different communication channels. While these methods help convey information to students and staff, they often lack efficiency, centralized management, and proper tracking mechanisms.

At present, most institutions do not have a centralized digital platform that manages the creation, approval, and distribution of circulars in a structured manner. Circulars are often sent through multiple communication sources such as email groups, messaging applications, or departmental notice boards. Because of this scattered approach, students and staff may miss important announcements or receive delayed information. Additionally, there is no proper system to verify whether the intended recipients have viewed or acknowledged the circulars, which creates communication gaps between the administration and the users.

Access to important academic information through these traditional methods is also limited in terms of reliability and accessibility. Students may not always visit notice

boards regularly, and emails or messages may get lost among other communications. As a result, important notifications related to examinations, academic schedules, seminars, or institutional policies may not reach all intended recipients on time. This lack of real-time updates and centralized access can lead to confusion, miscommunication, and inconvenience for both students and staff members.

Record management and circular tracking in the existing system are often handled manually or through basic documentation tools. Administrative staff must maintain records of circulars, approvals, and distributions manually, which can be time-consuming and prone to human errors. There is also limited capability for monitoring circular history, analysing communication effectiveness, or managing approval workflows efficiently. These limitations highlight the need for a more organized, automated, and technology-driven system to improve the efficiency and reliability of circular management in educational institutions.

1.3 DRAWBACKS OF EXISTING SYSTEM

The traditional circular distribution system suffers from several limitations:

- Lack of a centralized platform for managing and distributing circulars efficiently
- Important announcements may not reach all students and staff on time
- Dependence on physical notice boards which require users to be physically present
- Difficulty in tracking whether recipients have viewed or acknowledged the circular
- Manual preparation and distribution of circulars increases administrative workload
- Risk of miscommunication due to multiple communication channels such as emails and messaging groups
- Limited security and control over who can create, approve, or modify circulars
- Absence of an automated approval workflow for verifying circulars before publishing
- Difficulty in maintaining and retrieving historical records of previously issued circulars

- Time-consuming process for administrators to manage, update, and distribute circular information

These drawbacks result in inefficient communication, delays in information dissemination, and increased administrative effort within educational institutions.

1.4 PROPOSED SYSTEM

The proposed system, Department Circular Management System, is a cloud-based platform designed to overcome the limitations of the existing manual and semi-digital circular distribution methods used in educational institutions. The system is developed using the Salesforce platform, which provides a secure, scalable, and centralized environment for managing institutional announcements and circular communications. By digitalizing the circular management process, the system enables efficient creation, approval, and distribution of circulars to students and staff within the institution.

The system allows administrators to create circulars digitally through a centralized interface where details such as title, description, department, target audience, and attachments can be added. Once a circular is created, it follows an approval workflow where the Head of Department (HOD) reviews and approves or rejects the circular before it becomes visible to users. This approval process ensures that all circulars are verified before being published, improving accuracy and maintaining proper administrative control.

A key feature of the proposed system is the integration of Salesforce Experience Cloud, which provides a dedicated portal where students and staff can securely log in and view circulars relevant to them. Role-based access control ensures that users only see circulars intended for their category, such as students or staff. Automated email notifications inform users whenever a new circular is approved and published, ensuring timely communication across the institution.

The system includes features such as student acknowledgment of circulars, department-wise circular targeting, and centralized record management. Administrators can easily manage circular history, monitor approvals, and track user interactions within the system. By leveraging Salesforce automation tools such as approval processes,

workflows, and cloud-based access, the proposed system improves transparency, reduces manual effort, and provides a reliable digital communication platform for academic institutions

1.5 ADVANTAGES OF PROPOSED SYSTEM

- Provides a centralized digital platform for managing and distributing circulars within the institution
- Enables quick and efficient communication between administration, staff, and students
- Reduces dependency on physical notice boards and paper-based circular distribution
- Allows role-based access control so users can view only relevant circulars
- Automates the circular approval process through HOD verification before publishing
- Minimizes manual work and reduces administrative effort in managing announcements
- Provides real-time notifications and email alerts for newly approved circulars
- Maintains a secure and organized digital record of all circulars for future reference

SUMMARY

This chapter discussed the need for improving communication within educational institutions by replacing traditional circular distribution methods with a digital solution. The limitations of existing systems such as delayed information delivery, lack of centralized management, and difficulty in tracking circular distribution were identified. To address these issues, the proposed Department Circular Management System developed using the Salesforce platform provides a cloud-based solution that enables administrators to create circulars, implement an approval process through the Head of Department, and distribute approved circulars efficiently to students and staff. The system ensures secure access, role-based visibility, and centralized management of circular records, thereby improving transparency and communication efficiency within the institution

CHAPTER 2

SYSTEM ANALYSIS

2.1 IDENTIFICATION OF NEED

Educational institutions require an efficient and reliable communication system to ensure that important announcements and circulars reach students and staff in a timely manner. However, many institutions still rely on traditional communication methods such as physical notice boards, printed circulars, emails, or informal messaging platforms. These methods often lead to delays in information delivery, lack of centralized management, and difficulty in ensuring that all intended recipients receive the information. Additionally, the absence of a structured approval system may result in unverified or inconsistent communication being circulated within the institution.

There is a strong need for a centralized digital platform that can streamline the process of creating, approving, and distributing circulars within educational institutions. Such a system should provide secure access, role-based control, and real-time availability of circular information for students and staff. The identified need of this project is to develop a cloud-based circular management system using the Salesforce platform that improves communication efficiency, ensures proper approval workflows, and provides a centralized platform for managing institutional announcements and notifications.

2.2 FEASIBILITY STUDY

A feasibility study is an important phase in system analysis that helps determine whether the proposed system can be successfully developed and implemented within the given constraints. It evaluates the practicality, efficiency, and sustainability of the system before the actual development process begins. The feasibility study ensures that the project is realistic, cost-effective, technically achievable, and acceptable to users.

In this project, the feasibility study is conducted to analyze the proposed Department Circular Management System from three major perspectives: economic feasibility, operational feasibility, and technical feasibility. Economic feasibility examines whether the benefits of the system justify the cost involved in development and implementation.

Operational feasibility focuses on how effectively the system can function within the institutional environment and whether users such as administrators, staff, and students can easily adapt to the new system. Technical feasibility evaluates whether the required technologies, tools, and technical resources are available for developing and maintaining the system.

Performing this feasibility analysis, potential challenges and risks can be identified early in the development process, allowing appropriate solutions to be planned in advance. This study confirms that the proposed system is practical, reliable, and capable of improving communication efficiency within the institution. The system will provide a centralized platform for creating, approving, and distributing circulars effectively.

Three considerations involved in feasibility are:

- Economic Feasibility
- Operational Feasibility
- Technical Feasibility

2.2.1 Economic Feasibility

Economic feasibility evaluates whether the proposed system is financially viable and whether the expected benefits outweigh the development and operational costs. The proposed Department Circular Management System is economically feasible because it is developed using the Salesforce platform, which provides a cloud-based environment that reduces the need for additional hardware or infrastructure.

Since the system operates on a cloud platform, institutions do not need to invest heavily in physical servers or maintenance costs. The system reduces manual work involved in preparing, distributing, and managing circulars, thereby minimizing administrative effort and operational expenses. By automating communication and approval workflows, the system improves efficiency and reduces the possibility of errors. Therefore, the overall cost of implementing the system is justified by the long-term benefits of improved communication and administrative management.

2.2.2 Operational Feasibility

Operational feasibility refers to how effectively the proposed system can function within the existing operational environment of the institution. The Department Circular Management System is designed to be simple, user-friendly, and easy to use for

administrators, staff members, and students.

The system provides an intuitive interface where administrators can create circulars, submit them for approval, and publish them after verification by the Head of Department. Students and staff can easily access circulars through a centralized digital portal and view announcements relevant to them. The automation of circular creation, approval, and distribution significantly reduces manual workload and improves communication efficiency.

Since the system integrates with existing institutional processes and does not require major changes to current workflows, it can be easily adopted by users without operational difficulties.

2.2.3 Technical Feasibility

Technical feasibility evaluates whether the necessary technical resources and expertise are available to develop and maintain the proposed system. The Department Circular Management System is technically feasible because it is developed using the Salesforce cloud platform, which provides built-in tools for application development, data management, and system security.

Salesforce offers features such as custom objects, approval processes, role-based access control, and Experience Cloud portals, which simplify the development of enterprise-level applications. These tools support scalability, security, and efficient data management, ensuring reliable system performance.

The required software tools and development environment are readily available, and the platform provides extensive documentation and support for developers. Therefore, the proposed system meets all technical requirements and can be successfully implemented and maintained within the institutional environment.

2.3 SOFTWARE REQUIREMENT SPECIFICATIONS

Software Requirement Specification (SRS) defines the functional and non-functional requirements of the proposed system. It provides a clear description of how the system should operate and outlines the hardware and software required for successful implementation. The SRS acts as a guideline for system design, development, and testing, ensuring that the final system meets the requirements of users and the objectives of the project.

The software requirement specification for the proposed Department Circular Management System identifies the essential technologies, tools, and platforms required to develop a secure, scalable, and cloud-based application. The system is developed using the Salesforce platform, which provides various tools and services for application development, data management, automation, and user access management.

2.3.1 Software Requirements

The following software requirements are necessary for the development and execution of the proposed system:

- **Operating System:** Windows 10 / Windows 11 / Linux / macOS
- **Development Platform:** Salesforce Platform
- **Programming Environment:** Salesforce Lightning Platform
- **Automation Tools:** Salesforce Flow, Approval Process, Workflow Rules
- **Database Management System:** Salesforce Cloud Database
- **User Interface Framework:** Lightning Experience / Experience Cloud
- **Web Browser:** Google Chrome, Mozilla Firefox, Microsoft Edge
- **Development Tools**: Salesforce Developer Console, Visual Studio Code
- **API Testing Tool:** Postman
- **Version Control System:** Git and GitHub

These software tools support efficient development, configuration, deployment, and maintenance of the proposed circular management system.

2.3.2 Hardware Requirements

The following hardware requirements are necessary for the development and execution of the proposed system:

- **Device Type:** Laptop / Desktop Computer
- **Processor:** Intel Core i5 / AMD Ryzen 5 or equivalent
- **Processor Speed:** 2.0 GHz or higher
- **Installed RAM:** 8 GB (minimum 4 GB required)
- **System Type:** 64-bit Operating System, x64-based Processor
- **Storage:** Minimum 256 GB Hard Disk / SSD
- **Display Resolution:** 1366 × 768 pixels or higher

These hardware specifications ensure smooth system performance, efficient processing, and reliable execution of the Department Circular Management System.

2.4 SOFTWARE DESCRIPTION

2.4.1 Front End

The front end of the proposed system is developed using the tools provided in the Salesforce Platform such as Lightning Components and Lightning App Builder, which help in designing interactive and user-friendly interfaces. These tools allow developers to create responsive web pages without complex coding, making the system easier to maintain and modify.

The front end provides different interfaces for administrators, staff members, and students. Administrators can create and upload circulars, staff can view and acknowledge circulars related to their department, and students can access circulars issued by the institution. The interface is designed in a simple dashboard format so that users can easily navigate through the system.

Additionally, the front end ensures a responsive design so that the application can be accessed through various devices such as computers, tablets, and mobile phones. This improves accessibility and ensures that important academic circulars reach users quickly and efficiently.

2.4.2 Back End

The back end of the system is implemented using the Salesforce CRM platform, which provides a secure and scalable cloud environment for application development. Salesforce allows the system to manage circular data, user information, and approval processes efficiently.

Custom objects are created in Salesforce to store circular information such as circular title, description, department, date of issue, and approval status. Administrators can upload new circulars, and the system automatically distributes them to the appropriate users based on their roles and departments.

Salesforce also supports automation features such as workflow rules and approval processes, which help streamline the circular management process. This reduces manual

effort and ensures that circulars are distributed quickly and accurately.

2.4.3 Database

The system uses the Salesforce cloud database to store and manage all application data efficiently. In Salesforce, data is organized using objects, fields, and records, which function similarly to tables, columns, and rows in traditional databases. This structured approach enables easy storage, retrieval, and management of information related to circulars and users.

The database stores important information such as user details (Admin, HOD, staff, and students), department information, circular records, acknowledgement data, and approval history. Since the data is maintained in a cloud environment, it can be accessed securely from any location with proper authentication and authorization. This ensures flexibility and real-time availability of information for all users.

Additionally, the Salesforce database provides strong security features such as role-based access control, data encryption, and user authentication to protect sensitive information. It also supports automatic backups, data recovery, and high availability, ensuring that data is not lost and remains accessible at all times. The scalable nature of the cloud database allows the system to handle increasing data efficiently, making it reliable for long-term use in institutional environments.

SUMMARY

This chapter presents the analysis of the proposed Department Circular Management System developed using the Salesforce Platform, highlighting the need to digitize the process of managing and distributing academic circulars within an institution. It examines the feasibility of the system from economic.The chapter also outlines the required hardware and software resources and describes the technologies used in system development. Overall, the analysis confirms that the proposed system is practical, secure, and efficient, providing a strong foundation for the system design and implementation discussed in the next chapter.

CHAPTER 3

SYSTEM DESIGN

3.1 MODULE DESCRIPTION

A module description provides a concise overview of a module's purpose and functionality within a system. It outlines the key features and possible use cases to help users understand its role. In software development, a module description explains how the module interacts with other components of the system. In the Department Circular Management System developed using the Salesforce Platform, modules are designed to efficiently manage the creation, approval, and distribution of academic circulars.

The project consists of the following modules:

- ➢ Admin Module
- ➢ Circular Creation Module
- ➢ HOD Module
- ➢ Approval Workflow Module
- ➢ Staff Module
- ➢ Student Module
- ➢ Notification Module
- ➢ User Management Module
- ➢ Circular Tracking Module
- ➢ Report & Dashboard Module

3.1.1 Admin Module

The Admin Module acts as the main control center of the Department Circular Management System developed using the Salesforce Platform. It provides administrators with complete access to manage system activities such as creating and managing circulars, maintaining department information, and controlling user accounts including

HODs, staff, and students. The admin can update or remove user details and assign appropriate roles and permissions within the system. This module also allows the administrator to monitor circular approvals, track published circulars, and view system reports through dashboards. By managing these functions, the admin ensures smooth communication and efficient distribution of circulars within the institution.

3.1.2 Circular Creation Module

The Circular Creation Module is an important component of the Department Circular Management System developed using the Salesforce Platform. This module allows authorized users such as administrators to create and upload circulars in a digital format. Users can enter details such as the circular title, description, department name, and issue date, and also attach related documents if necessary. Once the circular is created, it is stored in the system and forwarded for approval before being published. The module helps organize circulars by department or category, making it easier for users to access relevant information. By maintaining digital records of circulars, this module improves efficiency, reduces manual work, and ensures that important institutional announcements are properly managed and distributed.

3.1.3 HOD Module

The HOD Module allows the Head of Department to review and manage circulars related to their respective department within the system developed using the Salesforce Platform. In this module, the HOD can view circulars that are submitted for departmental verification and ensure that the information is accurate and relevant. The HOD has the authority to approve, reject, or request modifications to the circular before it is officially released. This module helps maintain proper validation of departmental circulars and ensures that only authorized and verified information is distributed to staff and students. By involving the Head of Department in the approval process, the system improves accountability and ensures effective communication within the institution.

3.1.4 Approval Workflow Module

The Approval Workflow Module manages the review and approval of circulars before publication in the system. After a circular is created, it is forwarded to the appropriate authority such as the HOD or administrator for verification. The authorized

user reviews the circular details and can approve, reject, or request modifications if needed. Only approved circulars are released and made visible to staff and students. This ensures a proper verification process, maintains accuracy and reliability, and provides official authorization. The module also keeps records of approval activities to ensure transparency and improve accountability in the circular management process.

3.1.5 Report & Dashboard Module

The Report & Dashboard Module provides a visual overview of system activities using interactive dashboards. It displays key information such as total circulars, pending approvals, and recently published circulars. Administrators can monitor system performance and track circular distribution across departments. The module generates reports to analyze circular records and user activity. These reports support better decision-making and proper documentation. The dashboard also provides real-time insights to manage workflows efficiently. It helps track approval status and overall system usage. Additionally, users can generate detailed summaries of circular activities for effective management and record keeping.

3.1.6 Notification Module

The Notification Module keeps users updated about newly published circulars and important announcements. It sends alerts to administrators, HODs, staff, and students when circulars are created, approved, or released. Notifications ensure timely communication and help users access important information quickly. The module also provides reminders for pending approvals and new circulars. This improves communication, and ensures that no important updates are missed within the institution.

3.1.7 User Management Module

The User Management Module manages all user accounts in the system. It allows administrators to create, update, and manage users such as HODs, staff, and students. Each user is assigned roles and permissions to control access to system features. The module ensures secure login and authentication to protect system data. It also enables administrators to update user details, reset passwords, and deactivate accounts when required. Additionally, it maintains user records and helps track user activity, ensuring efficient and secure system management.

3.1.8 Staff Module

The Staff Module allows staff members to access and view circulars related to their department. Staff can log in to the system and stay updated with important academic and administrative announcements. The module provides an organized interface to view recent and past circulars easily. It ensures timely communication of departmental information. Staff can also track updates and respond quickly to important notices. Additionally, the module helps staff stay informed about institutional activities and supports better coordination with the administration. This module improves communication through a centralized and efficient platform.

3.1.9 Student Module

The Student Module allows students to access important circulars and announcements through the system. Students can log in and view circulars related to academics, exams, events, and other updates. The module provides a simple interface to read new and previous circulars easily. It ensures timely delivery of information without relying on notice boards. Students can stay updated and respond quickly to announcements. This module improves communication by providing a centralized platform for accessing official information.

3.2 DATA FLOW DIAGRAM

A Data Flow Diagram is a graphical representation that shows the flow of data through a system. It illustrates how data moves from one entity to another, how it is processed, and how it is stored. DFDs help in understanding the overall system functionality and interactions between external entities and internal processes

3.2.1 DATA FLOW DIAGRAM Level 0

Data Flow Diagram Level 0 provides a high-level overview of the system by showing it as a single process interacting with external entities. In the Campus Circular Management System, the main entities are Admin, HOD, and Student. The admin submits circulars, which are sent to the HOD for approval, and once approved, they are published for students. This ensures efficient management, verification, and distribution of circular information within the institution.

- The Admin interacts with the system by creating and submitting circular details. The admin is responsible for managing circular information and sending it to the system for further processing. The system stores the circular data and forwards it for approval. The admin can also update circular details, manage user information, and monitor the overall circular management process. In addition, the admin can access system reports and dashboard information to monitor circular activities, track approval status, and analyze the performance of the system.
- The Head of Department interacts with the system by reviewing the circular submitted for approval. The system sends the circular request to the HOD for verification. The HOD can either approve or reject the circular based on its validity and relevance. If the circular requires modifications, it can be sent back for correction before approval. Once the decision is made, the approval status is returned to the system, which determines whether the circular should be published.
- The Student interacts with the system by receiving and viewing newly published circulars. Once the circular is approved by the HOD, the system distributes the circular information to students through the platform. Students can log in to the system to read circular details, check important announcements, and stay informed about academic updates, events, or administrative notices issued by the institution.

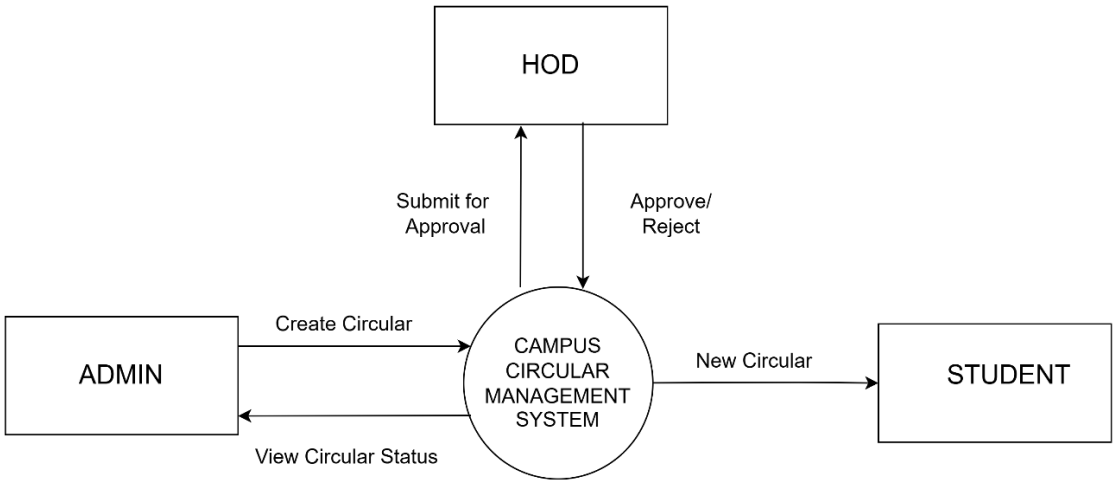

Figure 3.1 Data Flow Diagram Level 0

Figure 3.1 shows the Level 0 Data Flow Diagram, highlighting data flow, user interactions, and system boundaries.

3.2.2 DATA FLOW DIAGRAM Level 1

Data Flow Diagram Level 1 explains the internal working of the Campus Circular Management System. It divides the system into major processes and shows how data flows between users, system modules, and databases. In this system, the main processes include Circular Creation, Circular Review & Approval, Circular Publishing & Distribution, Viewer Tracking, and Reporting & Dashboard Management.

Create & Submit Circular

- The Admin creates a circular by entering details such as circular content, department and date.
- The system receives the circular information and marks the status as pending.
- The circular details are stored in the Circular Database (Pending Circulars).
- The system forwards the circular for review and approval by the Head of Department (HOD).Review & Verify Circular (HOD) .
- The HOD receives the circular review request from the system.
- The HOD checks the circular details to verify the correctness and relevance of the information.
- The HOD can either approve or reject the circular.
- If approved, the circular status is updated as approved and sent to the publishing process.
- If rejected, the system sends a rejection notification to the admin.

Publish & Distribute Circular

- Once the circular is approved, the system publishes the circular.
- The approved circular is stored in the Active Circular Database.
- The system distributes the circular notifications to staff and students.
- Staff and students receive the circular information through the system interface.

Track & Log Viewers

- When staff or students access the circular, the system records the view request.
- The system tracks which users have viewed the circular.
- Viewer information is stored in the Engagement Logs Database.
- This process helps monitor how many users have accessed the circular.

Log View Request (Detailed)

- When staff or students access the circular, the system records the view request.
- The system tracks which users have viewed the circular.
- Viewer information is stored in the Engagement Logs Database.
- This process helps monitor how many users have accessed the circular.

Aggregate & Report View Stats

- The system collects and processes the viewer log data.
- It calculates statistics such as view counts and engagement percentage.
- The system generates reports and dashboard data for monitoring circular activity.
- These reports are displayed in the User Interface for Admin, HOD, and staff monitoring.

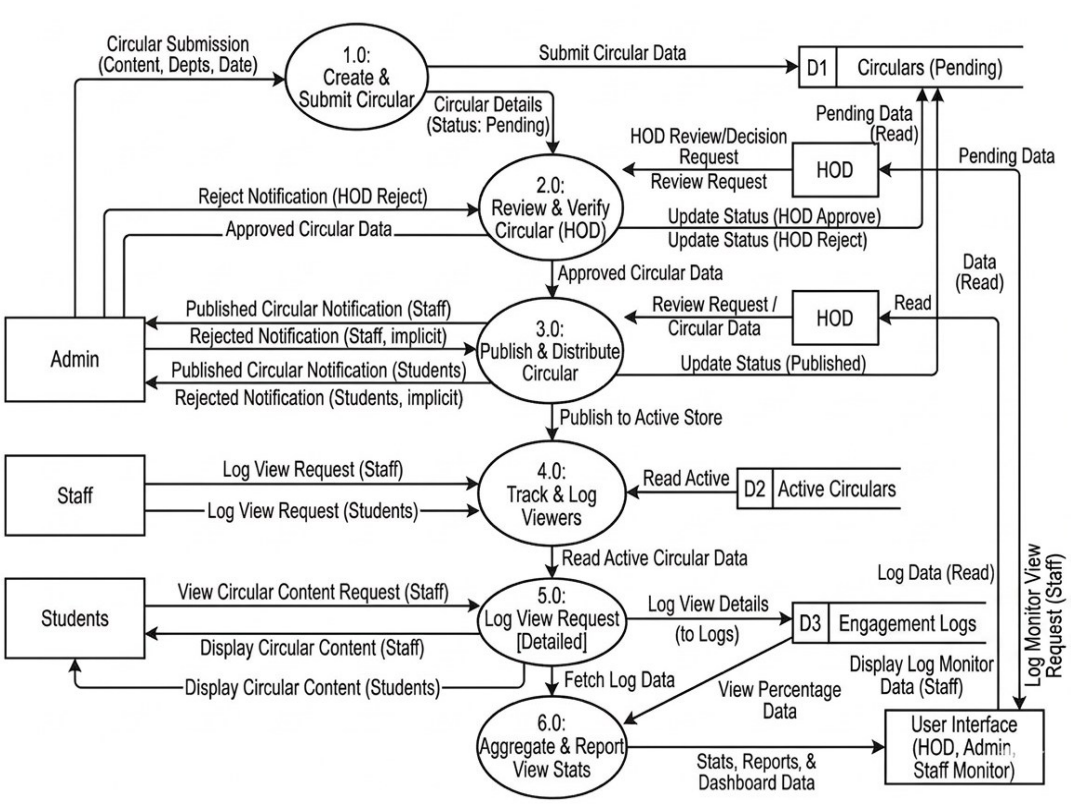

Figure 3.2 Data Flow Diagram Level 1

Figure 3.2 shows the Level 1 Data Flow Diagram, explaining internal processes and data flow from circular creation and approval to distribution, tracking, and reporting.

3.3 SYSTEM FLOW DIAGRAM

A System Flow Diagram represents the sequence of operations carried out within a system from the beginning of a process to its completion. It illustrates how users interact with the system, the decisions made during the process, and the actions executed by the system. Unlike a Data Flow Diagram, which mainly focuses on the movement of data, a System Flow Diagram emphasizes the logical order of processes, decision points, and system responses.

The System Flow Diagram of the Department Circular Management System using Salesforce demonstrates how circulars are created, approved, distributed, and acknowledged within the institution. It explains the interaction between administrators, the Head of the Department (HOD), staff members, and students.

The system flow begins when the Administrator logs into the system. After successful authentication, the administrator can create and post a new circular by entering the necessary information such as the circular title, description, attachments, and target audience.

Once the circular is created, it is automatically sent to the HOD for approval. The HOD reviews the content and decides whether to approve or reject it, ensuring that only verified circulars are published. If required, the circular can be sent back for modifications before approval.

At this stage, a decision point occurs:

- If the HOD approves the circular, the system proceeds to distribute the circular to all relevant staff members and students.
- If the HOD rejects the circular, the system returns it to the administrator along with the reason for rejection, allowing necessary modifications before reposting.

When the circular is approved, it becomes visible to students and staff within the system. Students can view the circular and acknowledge that they have read it.staff members can monitor the acknowledgements and responses from students.The system records these acknowledgements and maintains the information for administrative tracking and monitoring. Based on the collected data, the system generates reports and dashboards, enabling administrators and staff to analyze how many users have viewed or acknowledged the circular. This helps in measuring user engagement and ensuring that

important information reaches the intended audience. If users do not respond immediately, they can still access the circular later through the system dashboard. The process concludes when acknowledgements are recorded and the system updates the reports and dashboard with the latest information.

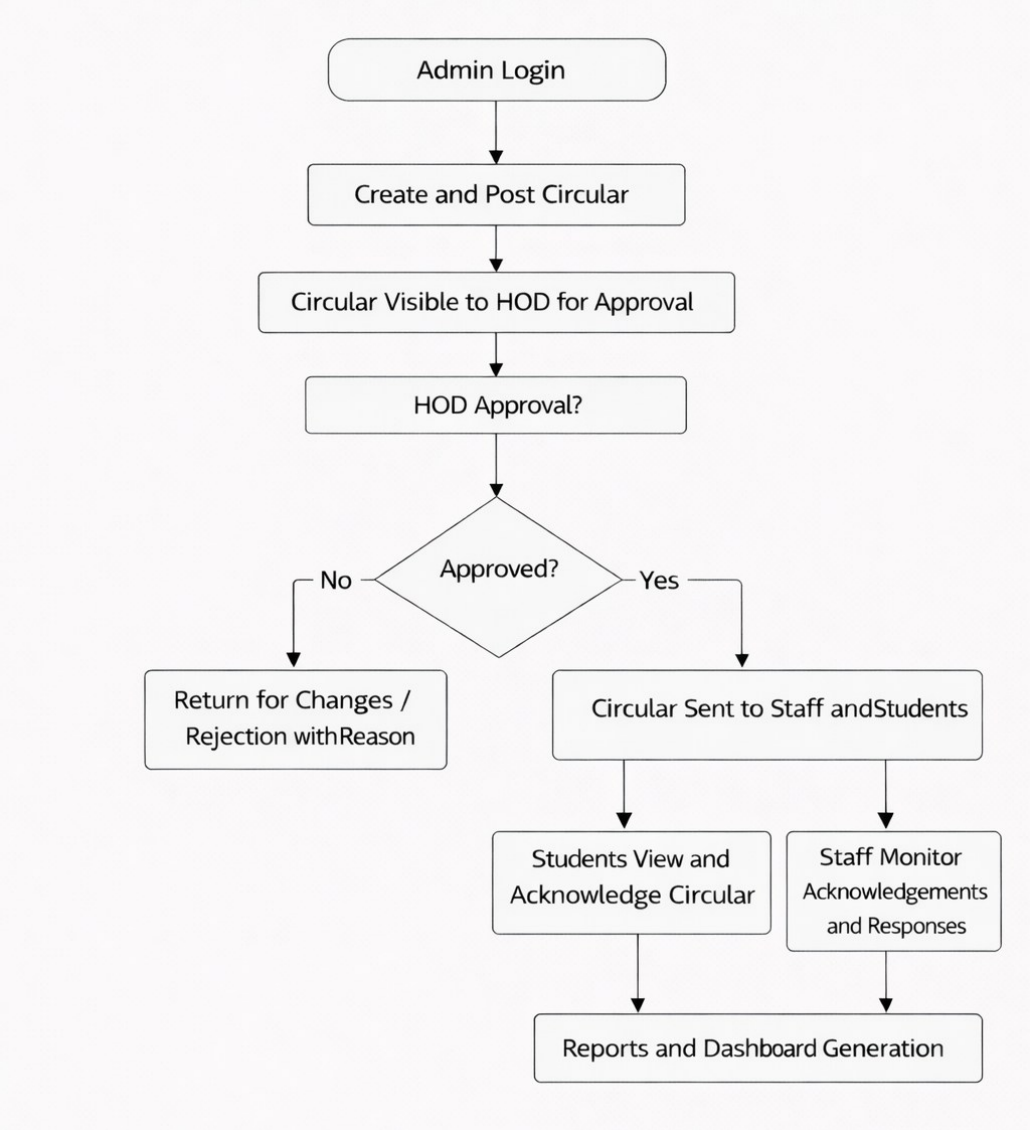

Figure 3.3 System flow Diagram

Figure 3.3 illustrates the System Flow Diagram of the Department Circular Management System using Salesforce. It clearly depicts the step-by-step sequence of operations, decision points, and interactions between administrators, HOD, staff, and students, providing a clear understanding of how circulars are created, approved, distributed, and monitored within the system.

3.4 INPUT DESIGN

Input design refers to the process of creating user interfaces that allow users to enter data into the system accurately and efficiently. A well-designed input interface helps reduce errors, improves usability, and ensures that the system collects the correct information required for processing. Effective input design focuses on simplicity, validation, clarity, and ease of use so that users can interact with the system without confusion. In the Department Circular Management System implemented using Salesforce, input forms are designed to allow administrators to create and manage circular announcements efficiently.

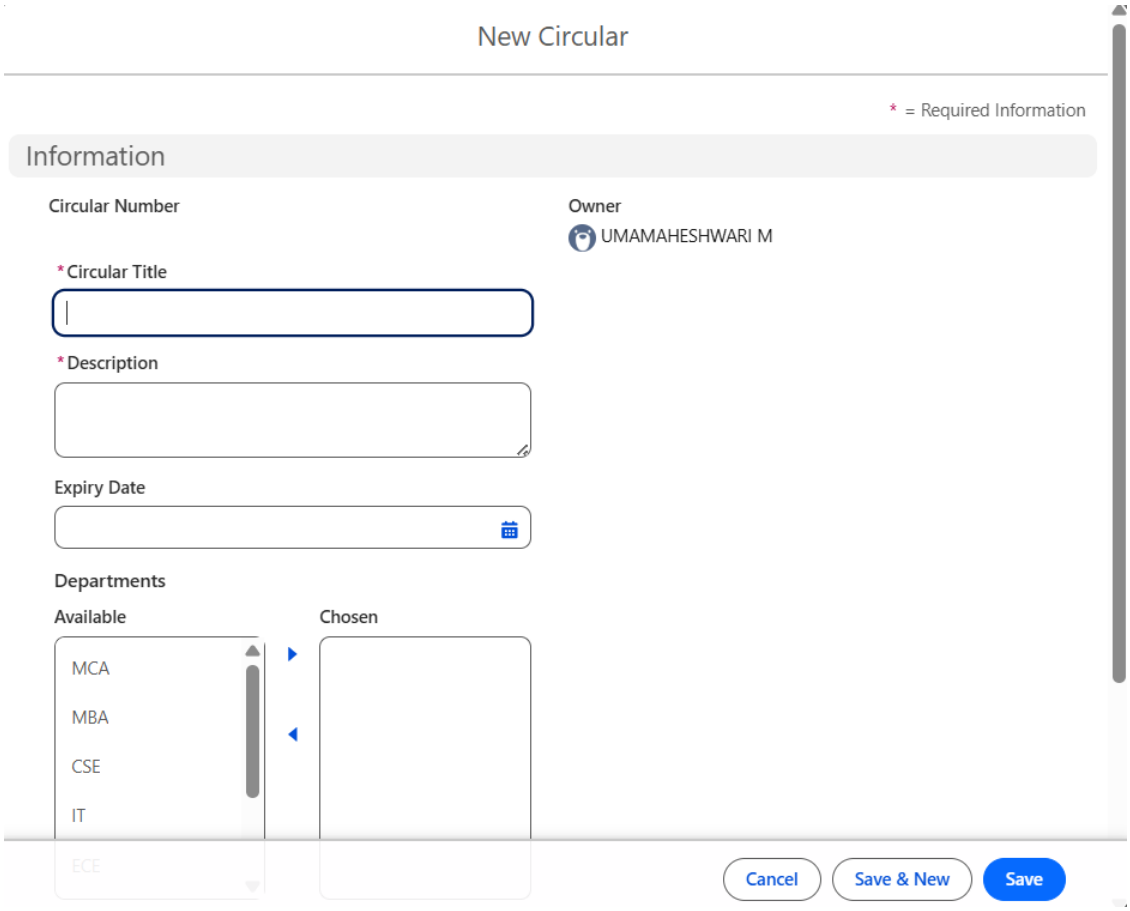

Figure 3.4 Circular Creation Form

Figure 3.4 shows the Circular Creation Form used by the administrator to enter circular details such as title, description, expiry date, and department. After entering the required information, the circular is saved and forwarded for approval, ensuring accurate data entry and efficient management.

3.5 OUTPUT DESIGN

Output design focuses on presenting processed information to users in a clear and meaningful format to support effective monitoring and decision-making. In the Department Circular Management System implemented using Salesforce, outputs are designed to be simple, organized, and easy to understand. This ensures that users can quickly access important information without confusion.

The system outputs include circular details, acknowledgement status, and report dashboards that help administrators and staff monitor circular distribution and student responses. These outputs display information such as issued circulars, viewed status, and acknowledgement counts, enabling users to track communication activities and analyze system performance efficiently. Additionally, graphical reports and dashboards provide visual insights, helping users make better decisions and manage circular activities more effectively.

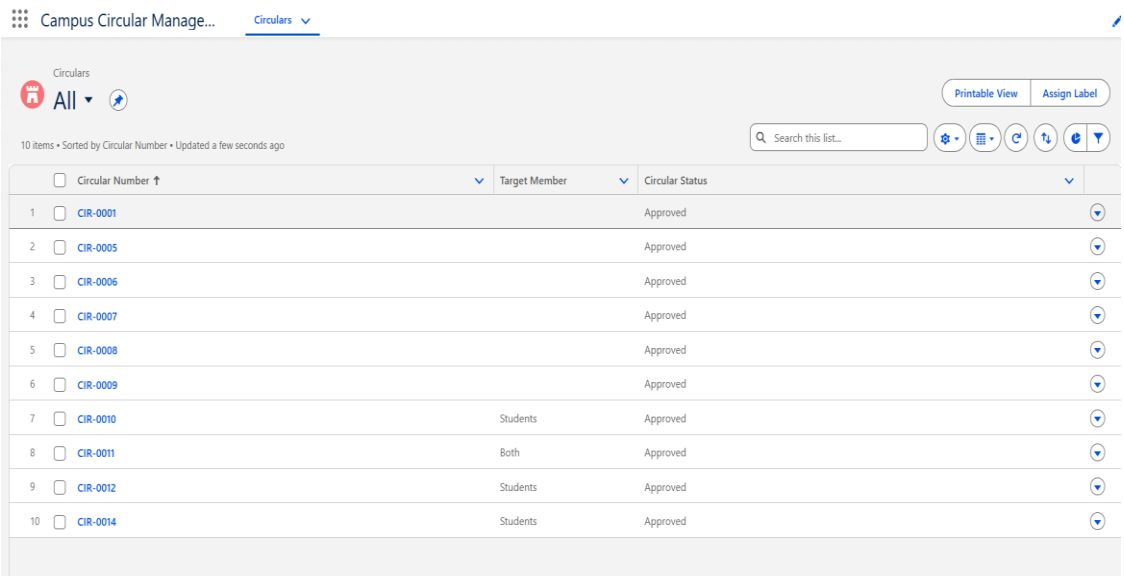

Figure 3.5 Circular List Output

Figure 3.5 shows the Circular List Output Screen displaying details such as Circular Number, Target Member, and Status. It allows users to view, search, and manage circular records, helping monitor approvals and track issued circulars efficiently.

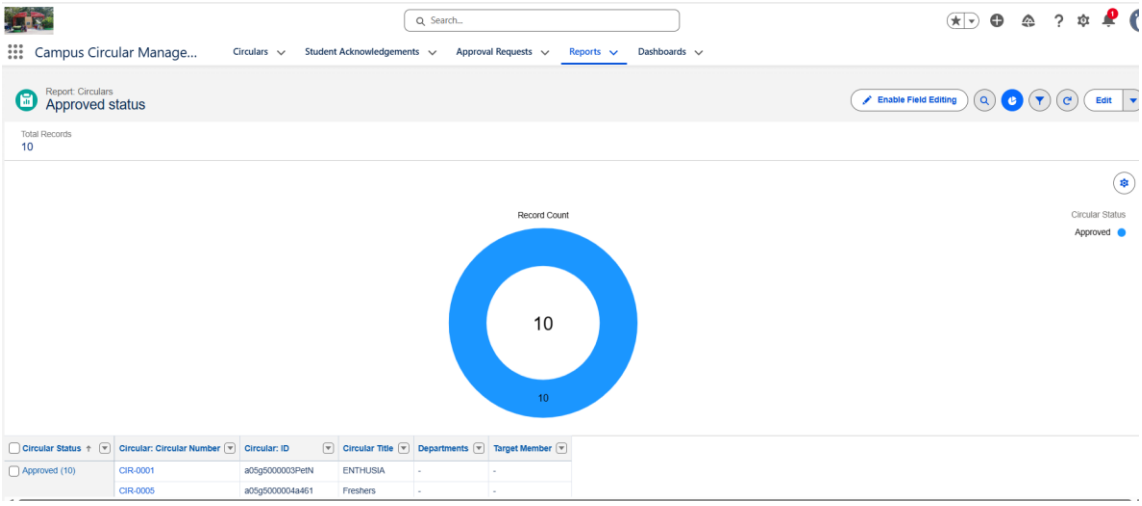

Figure 3.6 Approved Circular Report

Figure 3.6 Output design presents system information clearly to users. In the Department Circular Management System using Salesforce, outputs such as circular lists, reports, and dashboards help administrators monitor circular status and system activities.

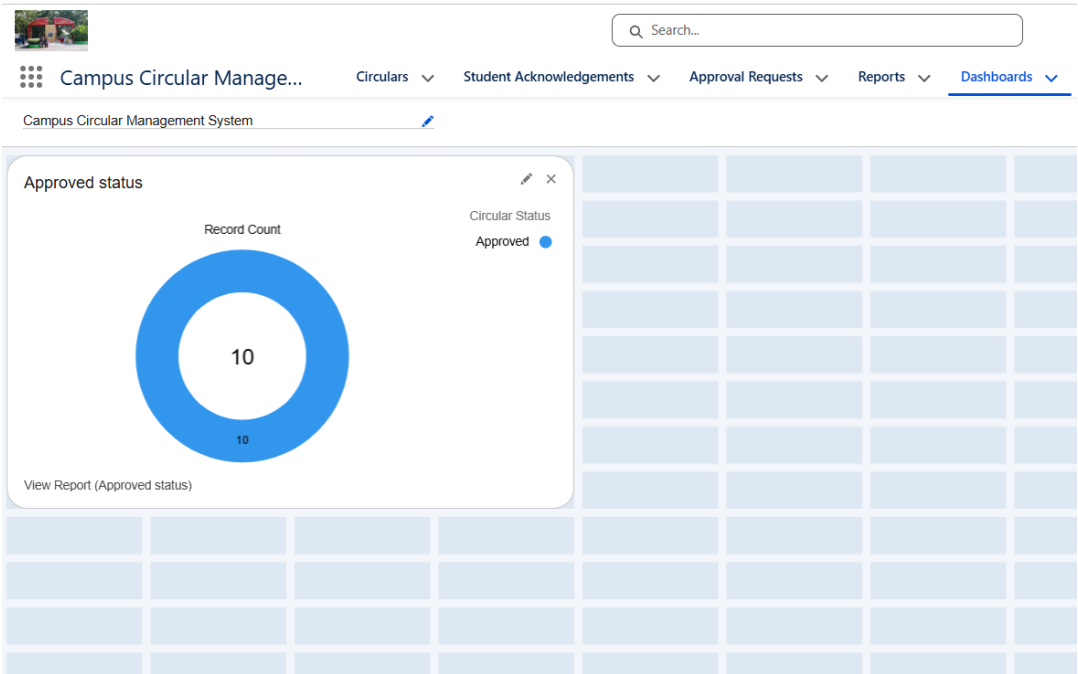

Figure 3.7 Dashboard

Figure 3.7 shows the Dashboard View of the Department Circular Management System implemented using Salesforce. It displays graphical charts showing the number of approved circulars and system activity, helping administrators quickly monitor and analyze circular management data.

3.6 ENTITY RELATIONSHIP DIAGRAM

An Entity Relationship Diagram is a conceptual representation of the data structure of a system. It illustrates the entities involved in the system, their attributes, and the relationships between them. ER diagrams help in understanding how data is organized, stored, and connected within the system. They provide a clear overview of database design and support efficient data management.

The Entity Relationship Diagram of the Department Circular Management System represents the main entities involved in managing circulars within the institution. The primary entities identified in the system include Admin, HOD, Staff, Student, Circular, Acknowledgement, and Report Dashboard. Each entity contains attributes that store relevant information and establish relationships with other entities.

Admin

The Admin entity represents the administrator who manages the circular management system. The attributes of this entity include Admin_ID, Admin_Name, Email, Password, and Role. The admin is responsible for creating and managing circulars in the system. The admin also manages the circular records and oversees system operations.

HOD

The Head of Department (HOD) entity is responsible for reviewing and approving circulars before they are distributed. The attributes of this entity include HOD_ID, HOD_Name, Department, and Email. The HOD is connected to the circular entity through the Approves relationship, indicating that the HOD has the authority to approve or reject circulars created by the admin.

Staff

The Staff entity represents faculty members who receive circular notifications and monitor student acknowledgements. The attributes of this entity include Staff_ID, Staff_Name, Department, and Email. Staff members are related to the circular entity through the Sends relationship, which indicates that circulars are communicated to staff members.

Student

The Student entity represents the students who receive circular notifications and acknowledge them after reading. The attributes of this entity include Student_ID, Roll_Number, Department, and Email. Students receive circulars through the Receives relationship with the circular entity.

Circular

The Circular entity is the central component of the system that stores information about circular announcements. The attributes include Circular_ID, Description, Created_Date, and Status. Circulars are created by the admin, approved by the HOD, and distributed to staff and students.

Acknowledgement

The Acknowledgement entity records whether a student has viewed or acknowledged a circular. The attributes include Ack_ID and Viewed_Status. This entity is connected to the student through the Acknowledges relationship, indicating that students confirm they have read the circular.

Report Dashboard

The Report Dashboard entity represents system-generated analytics and monitoring information. The attributes include Report_ID, Total_Circulars, and Viewed_Count. The circular entity generates reports that help administrators and staff track how many circulars have been issued and how many users have acknowledged them. These reports provide insights into user engagement and communication effectiveness within the institution.

The relationships between these entities define how the system operates. The Admin manages circulars, the HOD approves circulars, the system sends circulars to staff and students, and students acknowledge the circular after viewing it. The system then generates reports and dashboard analytics based on the circular data and acknowledgement records. This integrated workflow ensures efficient communication, proper tracking, and improved decision-making.

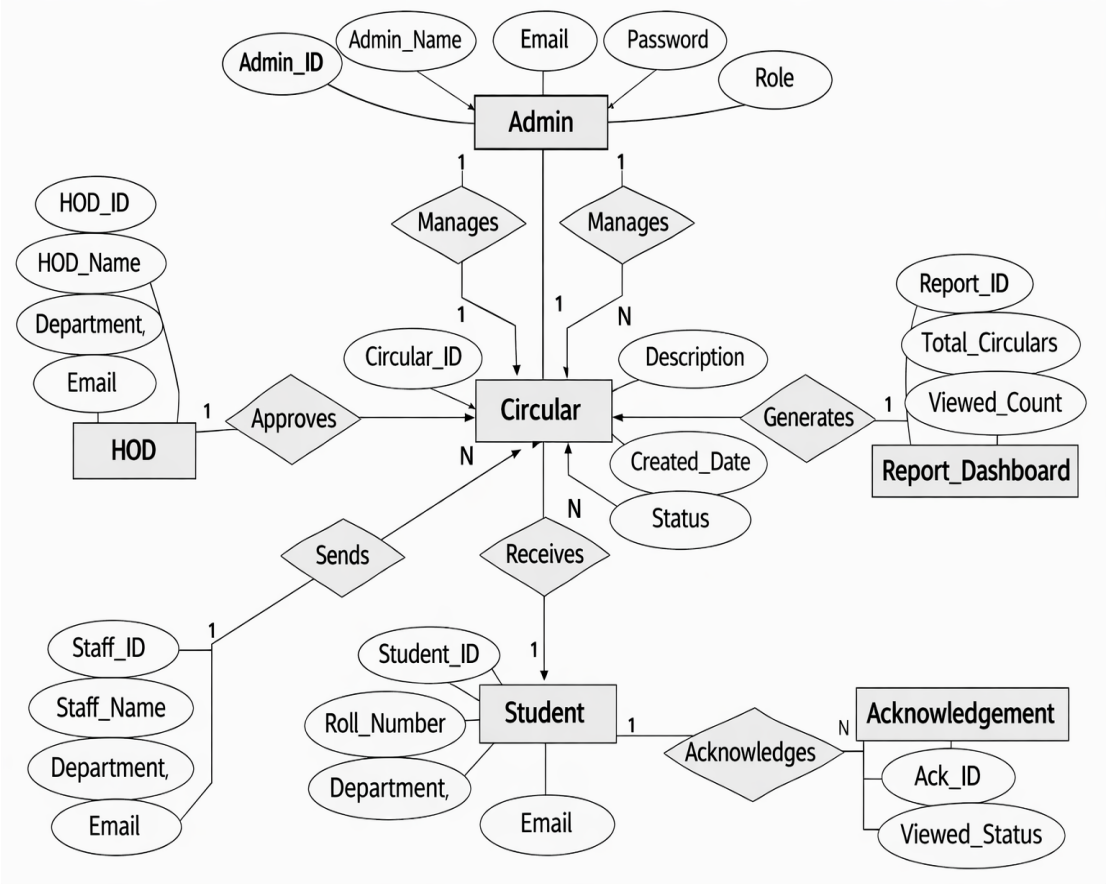

Figure 3.8 Entity Relationship Diagram

Figure 3.8 illustrates the Entity Relationship Diagram of the Department Circular Management System. It shows the entities, attributes, and relationships involved in managing circular creation, approval, distribution, acknowledgement, and reporting within the system.

3.7 DATABASE DESIGN

Database design for the Department Circular Management System using Salesforce focuses on organizing and managing system data efficiently to support communication and monitoring within the institution. Salesforce uses a relational data structure where objects act as tables and records represent stored information. This structure allows the system to store and manage data related to circular announcements, users, acknowledgements, and reports effectively.

In the proposed system, Salesforce objects store important information such as Admin details, HOD information, Staff records, Student details, Circular announcements, and Acknowledgement status. The main objects in the system include Admin, HOD,

Staff, Student, Circular, Acknowledgement, and Report Dashboard. Objects organize data to store, retrieve, update, and manage circular information and user responses.

Data Integration

Data integration in the Department Circular Management System combines information from different modules such as circular creation, approval, distribution, acknowledgement tracking, and reporting. This integration allows administrators and staff members to manage all circular-related activities within a single system.

By integrating these modules, the system enables efficient tracking of circular announcements, monitoring of student acknowledgements, and generation of reports through dashboards. This unified data management improves communication efficiency and provides better visibility of system activities.

Data Independence

Salesforce provides strong data independence through its object-based architecture. Each object in the system, such as Student, Staff, Circular, and Acknowledgement, can operate independently while maintaining relationships with other objects.

This flexibility allows administrators to add new fields or modify existing data structures without affecting other parts of the system. For example, additional attributes can be added to circular records or student information without impacting other objects. This improves scalability and simplifies system maintenance.

Data Security

Data security is an essential aspect of the system database design. The Department Circular Management System uses Salesforce security mechanisms to protect sensitive information and ensure controlled system access.

Salesforce implements role-based access control, authentication, and permission settings to ensure that only authorized users can access specific system features. Different roles such as Admin, HOD, Staff, and Student have different levels of access based on their responsibilities. These security features help protect system data, prevent unauthorized access, and maintain data integrity.

3.8 TABLE DESIGN

Database table design organizes data in a structured format for efficient storage, retrieval, and management. In the Department Circular Management System using Salesforce, the Circular table stores all circular-related information and ensures data consistency. It includes fields for circular details such as title, description, dates, and target audience. Proper constraints and field types help maintain data accuracy and prevent invalid entries.

This design supports efficient handling, updating, and tracking of circular records within the system. Additionally, it enables better organization of data, improves system performance, and simplifies data management for administrators.

Table 3.1 Circular Table Design

FIELD NAME	DATA TYPE	SIZE	CONSTRAINTS
Circular ID	Auto Number	10	PRIMARY KEY
Circular Title	Text	100	NOT NULL
Description	Long Text Area	255	NOT NULL
Created Date	Date	-	NOT NULL
Expiry Date	Date	-	NOT NULL
Department	Picklist	-	NOT NULL
Target Member	Picklist	-	NOT NULL
Circular Status	Picklist	-	NOT NULL

Table 3.1 shows the Circular Table Design with fields like ID, Title, Description, Dates, Department, Target Member, and Status. The primary key and constraints ensure valid data and efficient record management. The table stores circular information in a structured format, making it easy to manage. It supports quick retrieval and updating of records. It also helps track circular status and ensures consistency and accuracy of data within the system.

Table 3.2 Acknowledgement Table Design

FIELD NAME	DATA TYPE	SIZE	CONSTRAINTS
Acknowledgement ID	Auto Number	10	PRIMARY KEY
Circular ID	Lookup (Circular)	-	NOT NULL
Student Name	Lookup (Student)	-	NOT NULL
Viewed Status	Picklist	-	NOT NULL
Viewed Date	Date	-	NOT NULL

Table 3.2 shows the Acknowledgement Table Design with fields like Acknowledgement ID, Circular ID, Student Name, Viewed Status, and Viewed Date. The primary key, lookup fields, and constraints ensure proper data tracking and relationships. This table helps record whether a student has viewed a circular. It supports efficient monitoring of user acknowledgements and improves tracking of circular communication within the system.

SUMMARY

This Chapter explains the overall system design of the Department Circular Management System using Salesforce. It describes the main components of the system such as Admin, HOD, Staff, Student, Circular management, Acknowledgement tracking, and Report Dashboard. The chapter includes the System Flow Diagram, which shows the process of creating circulars, approving them by the HOD, sending them to staff and students, and tracking acknowledgements. It also presents the Entity Relationship Diagram, which illustrates the entities, attributes, and relationships used to store and manage circular and user data. The design also supports role-based access, ensuring that each user performs tasks based on their responsibilities within the system. Overall, this chapter provides the structural design and workflow of the system that supports the implementation discussed in the next chapter.

CHAPTER 4

IMPLEMENTATION

4.1 CODE DESCRIPTION

Code description plays a vital role in explaining the functionality and purpose of the system used in the Department Circular Management System. The project is implemented using Salesforce, which is used for managing data, workflows, reports, and dashboards within the system. Each module is carefully configured to ensure clarity, proper data handling, and easy system maintenance. This helps both developers and administrators understand system behavior and manage operations efficiently.

Instead of traditional programming code, Salesforce uses configuration-based development such as custom objects, fields, page layouts, validation rules, reports, and dashboards to implement system functionalities. These components manage operations such as circular creation, approval by the HOD, distribution to staff and students, acknowledgement tracking, and report generation. This structured approach improves efficiency, reduces manual effort, and ensures smooth system functioning.

Proper code description and configuration documentation make it easier to troubleshoot issues and perform future enhancements. It also helps new users and developers understand the system quickly, reducing learning time. This approach supports scalability, improves system reliability, and ensures long-term maintainability of the application.

4.2 STANDARDIZATION OF THE CODING

Coding standards are followed throughout the development of the Department Circular Management System to maintain consistency, readability, and proper system organization. The project is implemented using Salesforce, where standard configuration practices such as meaningful field names, clear object structures, proper page layouts, and organized reports and dashboards are maintained across the system. This ensures that all components are easy to understand and manage.

Following these standardized practices ensures that the system remains easy to update, maintain, and scale over time. Salesforce features such as custom objects, validation rules, reports, and dashboards are organized systematically to support circular creation, approval processes, acknowledgement tracking, and report generation. Additionally, consistent naming conventions and structured configurations reduce errors and improve development efficiency. This approach enhances system reliability, simplifies troubleshooting, and supports future enhancements of the application.

4.3 ERROR HANDLING

Error handling is an important part of the system implementation to ensure smooth operation and a better user experience. In the Department Circular Management System implemented using Salesforce, errors may occur due to invalid user inputs, missing data, permission issues, or incorrect field values during circular creation and updates. Identifying and handling these errors properly helps prevent system failures and ensures uninterrupted system usage.

To handle such situations, Salesforce provides built-in mechanisms such as validation rules, required fields, and permission settings. These features help prevent incorrect data entry and ensure that only valid information is stored in the system. When an error occurs, the system displays clear error messages to guide users in correcting the issue. Additionally, role-based access control ensures that users can only perform authorized actions, reducing security risks. This approach improves system stability, enhances data accuracy, and ensures reliable and secure system performance.

Furthermore, proper error handling helps in identifying issues quickly during system operation and simplifies debugging and maintenance. It also enhances user confidence by providing clear guidance when mistakes occur. By maintaining consistent validation and error reporting, the system ensures smooth workflow execution and reliable circular management within the institution.

Example 1:

If the administrator tries to create a circular without entering required fields such as Circular Title or Description, the system displays an error message and prevents the record from being saved.

Example message:

"Please fill in all required fields before saving the circular."

Example 2:

If the Expiry Date is earlier than the Created Date, the system displays a validation error**.**

Example message:

"Expiry Date must be greater than the Created Date."

Example 3:

If the administrator attempts to create a circular with the same Circular Number that already exists, the system prevents duplicate entries using validation rules.

Example message:

"This circular number already exists."

Example 4:

If a user without proper permission attempts to edit or delete a circular, the system restricts the action.

Example message:

"You do not have permission to perform this action."

SUMMARY

This chapter describes the implementation details of the Department Circular Management System developed using Salesforce, including system configuration, module implementation, and the importance of standardized development practices. It explains the use of validation rules and permission settings for effective error handling, and highlights how modules such as circular creation, approval workflow, user management, and reporting are integrated for efficient operation. Proper organization of objects, fields, reports, and dashboards ensures system reliability, security, and easy maintenance, while automation features reduce manual effort and improve performance. Overall, this chapter provides a clear understanding of system functionality and prepares the system for testing, which is discussed in the next chapter.

CHAPTER 5

TESTING AND RESULTS

5.1 TESTING

Testing is an important phase in the software development life cycle that ensures the reliability, correctness, and performance of the Department Circular Management System. This chapter explains the testing methods used to verify that the developed system meets the required functionalities and operates correctly under different conditions. Testing was performed after the system configuration and implementation were completed, and the results were documented using appropriate test cases and outcomes.

The system was tested at different levels to detect and remove errors, verify module functionality, and ensure smooth interaction between system components. Each module was tested individually and later as part of the integrated system. The testing techniques used in this project include:

- Unit Testing
- Integration Testing
- Validation Testing

5.1.1 UNIT TESTING

Unit testing focuses on testing individual modules of the system independently to ensure that each component works correctly. In this project, unit testing was conducted during the development phase. Each module such as user login, circular creation, circular approval, circular viewing, and report generation was tested separately to verify system functionality and detect possible errors.

CASE 1

Module: User Login Module

Test Type: Authentication Verification

Input: Valid Username and Password

Output: User Dashboard Opens

TEST
Input: Username and Password

Output: Dashboard Page Displayed

Analysis: The actual output matches the expected output.

Result: Pass

CASE 2
Module: Circular Creation Module

Test Type: Input Validation

Input: Circular Title, Description, Expiry Date, Department

Output: Circular Created Successfully

TEST
Input: Valid Circular Details

Output: Circular Saved Successfully

Analysis: Circular record stored correctly in the system.

Result: **Pass**

CASE 3
Module: Circular Approval Module

Test Type: Approval Process

Input: Circular Submitted for Approval

Output: Circular Approved by HOD

TEST
Input: Click “Approve Circular”

Output: Circular Status Updated to Approved

Analysis: Approval process works correctly.

Result: **Pass**

CASE 4
Module: Circular List Module

Test Type: Circular Viewing

Input: Circular Records in System

Output: Circular List Displayed

TEST
Input: Open Circular List Page

Output: Circular Details Displayed in Table

Analysis: Circular list displayed correctly.

Result: Pass

CASE 5

Module: Report Dashboard Module

Test Type: Report Generation

Input: Circular Records Data

Output: Circular Report Displayed

TEST
Input: Open Dashboard Report

Output: Chart Showing Circular Status

Analysis: Report generated and displayed correctly.

Result: **Pass**

5.1.2 INTEGRATION TESTING

Integration testing verifies the interaction between multiple modules to ensure that data flows correctly throughout the system. In the Department Circular Management System implemented using Salesforce, it ensures that modules such as circular creation, approval, viewing, and reporting work together properly. This confirms that individually tested components function correctly when combined.

During integration testing, the flow of data between modules is carefully checked to ensure accuracy and consistency. For example, circulars created by the admin must correctly move to the approval stage and then be visible to users after approval. This ensures smooth communication between system components.

Integration testing also helps identify and resolve issues related to module interaction, such as data transfer errors or delays. It improves system performance, reliability, and ensures seamless coordination between all modules for efficient system operation.

CASE 1

Module: Admin + Circular Creation Module

Test Type: Circular Creation and Storage

Input: Create Circular → Save Record

Output: Circular Stored in System

TEST

Input: Enter circular details and save

Output: Circular appears in circular list

Analysis: Circular creation and storage modules work correctly.

Result: **Pass**

CASE 2

Module: Circular Creation + HOD Approval Module

Test Type: Approval Workflow

Input: Submit Circular for Approval

Output: Circular Status Updated

TEST

Input: HOD approves circular

Output: Circular status changed to "Approved"

Analysis: Circular approval workflow functions correctly.

Result: Pass

CASE 3

Module: Circular + Student Module

Test Type: Circular Distribution

Input: Approved Circular

Output: Circular Visible to Students

TEST

Input: Student logs in and views circular

Output: Circular displayed in circular list

Analysis: Circular distribution to students works correctly.

Result: Pass

CASE 4

Module: Circular + Dashboard Report Module

Test Type: Report Generation

Input: Circular Records Data

Output: Dashboard Charts Displayed

TEST

Input: Open dashboard report

Output: Circular statistics displayed in charts

Analysis: Dashboard report generated correctly.

Result: Pass

5.1.3 VALIDATION TESTING

Validation testing ensures that the Department Circular Management System meets user requirements and performs tasks according to institutional needs. It verifies that the system functions correctly in real-world scenarios and that all modules operate according to defined requirements. The system was tested to confirm that circular management, approval processes, and reporting features work as expected.

Additionally, validation testing ensures accurate results and a smooth user experience by aligning system functionalities with user expectations and institutional workflows. It also helps identify gaps before deployment, ensuring the system is efficient, user-friendly, and capable of supporting effective communication within the institution.

CASE 1

Module: Circular Creation Module

Test Type: Field Validation

Input: Missing Circular Title

Output: Validation Error Message

TEST

Input: Save circular without entering title

Output: Error message displayed

Analysis: Required field validation works correctly.

Result: Pass

CASE 2

Module: Circular Expiry Validation

Test Type: Date Validation

Input: Expiry Date Earlier than Created Date

Output: Validation Error Message

TEST

Input: Expiry date earlier than created date

Output: Error message displayed

Analysis: Date validation rules enforced correctly.

Result: Pass

CASE 3

Module: Circular Approval Module

Test Type: Approval Validation

Input: Circular submitted for approval

Output: Circular status updated after approval

TEST

Input: HOD approves circular

Output: Circular marked as "Approved"

Analysis: Approval process works according to requirements.

Result: Pass

SUMMARY

This chapter presents the testing and results of the Department Circular Management System implemented using Salesforce. Unit testing, integration testing, and validation testing were conducted to verify individual module functionality, ensure proper interaction between system components, and confirm that the system meets user requirements. Major modules such as user login, circular creation, circular approval, circular viewing, and dashboard reports were tested successfully, demonstrating that the system operates correctly and is ready for deployment.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

The Department Circular Management System, developed using Salesforce, provides an efficient solution for managing and distributing circular announcements within an institution. The system digitalizes the traditional circular communication process by enabling administrators to create circulars, obtain approvals from the Head of the Department (HOD), and distribute them to staff and students through a centralized platform.

The system improves communication efficiency by allowing users to easily access circular information, track acknowledgement status, and monitor circular activities through reports and dashboards. Features such as circular creation, approval workflows, acknowledgement tracking, and dashboard reports help administrators manage institutional communication more effectively and ensure that important announcements reach the intended users.

Overall, the project demonstrates how cloud-based technologies like Salesforce can be used to develop a scalable and reliable circular management system. By improving transparency, accessibility, and monitoring of circular distribution, the system enhances institutional communication and provides a modern digital solution for managing academic and administrative announcements.

6.2 FUTURE ENHANCEMENTS

Future enhancements to the Department Circular Management System can further improve its efficiency, accessibility, and overall functionality. Some potential improvements include:

Mobile Application Integration

A dedicated mobile application can be developed to allow students, staff, and

administrators to access circulars easily through their smartphones. Mobile notifications can alert users instantly when a new circular is published or approved.

Email and Notification Alerts

Future versions of the system can integrate automatic email and notification alerts to inform users whenever a new circular is issued or updated. This feature will ensure that important announcements reach users quickly and improve communication effectiveness.

Additionally, this enhancement can include customizable notification preferences, allowing users to receive alerts based on their roles or departments. Real-time notifications through mobile or web platforms can further improve user engagement and ensure that no important updates are missed, enhancing overall communication efficiency.

Advanced Analytics and Reporting

Enhanced reporting and analytics features can be added to provide detailed insights into circular distribution and acknowledgement rates. Advanced dashboards can help administrators monitor user engagement and track communication efficiency within the institution. These features can also support data-driven decision-making and improve overall system performance. They help identify trends and patterns, enabling better planning and management of institutional communication.

Document Attachment and File Management

Future enhancements may include improved document management features that allow administrators to attach files such as PDFs, images, or official notices to circulars. This will make circular communication more informative and organized.

Multi-Department and Role-Based Access Expansion

The system can be expanded to support multiple departments and advanced role-based access controls. This will allow different departments to manage their own circulars while maintaining centralized monitoring by administrators.

By incorporating these enhancements, the Department Circular Management System using Salesforce can evolve into a more intelligent, efficient, and scalable communication platform for managing institutional announcements

APPENDICES

A.SAMPLE CODE

BACKEND FILE

Circular_status

```
<?xml version="1.0" encoding="UTF-8"?>
<CustomField xmlns="http://soap.sforce.com/2006/04/metadata">
   <fullName>Circular_Status__c</fullName>
   <label>Circular Status</label>
   <required>false</required>
   <trackTrending>false</trackTrending>
   <type>Picklist</type>
   <valueSet>
      <restricted>true</restricted>
      <valueSetDefinition>
         <sorted>false</sorted>
         <value>
            <fullName>Draft</fullName>
            <default>true</default>
            <label>Draft</label>
         </value>
         <value>
            <fullName>Pending HOD Approval</fullName>
            <default>false</default>
            <label>Pending HOD Approval</label>
         </value>
         <value>
            <fullName>Approved</fullName>
            <default>false</default>
            <label>Approved</label>
         </value>
         <value>
            <fullName>Rejected</fullName>
            <default>false</default>
            <label>Rejected</label>
         </value>
      </valueSetDefinition>
   </valueSet>
</CustomField>
```

Circular_Title


```
<?xml version="1.0" encoding="UTF-8"?>
<CustomField xmlns="http://soap.sforce.com/2006/04/metadata">
   <fullName>Circular_Title__c</fullName>
   <externalId>false</externalId>
   <label>Circular Title</label>
   <length>100</length>
   <required>true</required>
   <trackTrending>false</trackTrending>
   <type>Text</type>
   <unique>false</unique>
</CustomField>
```


Department


```
<?xml version="1.0" encoding="UTF-8"?>
<CustomField xmlns="http://soap.sforce.com/2006/04/metadata">
   <fullName>Departments__c</fullName>
   <label>Departments</label>
   <required>false</required>
   <trackTrending>false</trackTrending>
   <type>MultiselectPicklist</type>
   <valueSet>
      <restricted>true</restricted>
      <valueSetDefinition>
         <sorted>false</sorted>
         <value>
            <fullName>MCA</fullName>
            <default>false</default>
            <label>MCA</label>
         </value>
         <value>
            <fullName>MBA</fullName>
            <default>false</default>
            <label>MBA</label>
         </value>
         <value>
            <fullName>CSE</fullName>
            <default>false</default>
            <label>CSE</label>
         </value>
         <value>
            <fullName>IT</fullName>
            <default>false</default>
```

```
            <label>IT</label>
          </value>
          <value>
            <fullName>ECE</fullName>
            <default>false</default>
            <label>ECE</label>
          </value>
          <value>
            <fullName>EEE</fullName>
            <default>false</default>
            <label>EEE</label>
          </value>
          <value>
            <fullName>MECH</fullName>
            <default>false</default>
            <label>MECH</label>
          </value>
          <value>
            <fullName>MSC</fullName>
            <default>false</default>
            <label>MSC</label>
          </value>
        </valueSetDefinition>
    </valueSet>
    <visibleLines>5</visibleLines>
</CustomField>
```


Description


```
<?xml version="1.0" encoding="UTF-8"?>
<CustomField xmlns="http://soap.sforce.com/2006/04/metadata">
    <fullName>Description__c</fullName>
    <label>Description</label>
    <required>true</required>
    <trackTrending>false</trackTrending>
    <type>TextArea</type>
</CustomField>
```


Expiry Date


```
<?xml version="1.0" encoding="UTF-8"?>
<CustomField xmlns="http://soap.sforce.com/2006/04/metadata">
    <fullName>Expiry_Date__c</fullName>
    <label>Expiry Date</label>
    <required>false</required>
```

```
    <trackTrending>false</trackTrending>
    <type>Date</type>
</CustomField>
```


Target Member


```
<?xml version="1.0" encoding="UTF-8"?>
<CustomField xmlns="http://soap.sforce.com/2006/04/metadata">
    <fullName>Target_Member__c</fullName>
    <label>Target Member</label>
    <required>false</required>
    <trackTrending>false</trackTrending>
    <type>Picklist</type>
    <valueSet>
        <valueSetDefinition>
            <sorted>false</sorted>
            <value>
                <fullName>Staffs</fullName>
                <default>false</default>
                <label>Staffs</label>
            </value>
            <value>
                <fullName>Students</fullName>
                <default>false</default>
                <label>Students</label>
            </value>
        </valueSetDefinition>
    </valueSet>
</CustomField>
```


List View


```
<?xml version="1.0" encoding="UTF-8"?>
<ListView xmlns="http://soap.sforce.com/2006/04/metadata">
    <fullName>All</fullName>
    <columns>NAME</columns>
    <columns>Target_Member__c</columns>
    <columns>Circular_Status__c</columns>
    <filterScope>Everything</filterScope>
    <label>All</label>
</ListView>
```


Date Validation


```
<?xml version="1.0" encoding="UTF-8"?>
<ValidationRule xmlns="http://soap.sforce.com/2006/04/metadata">
```

```
    <fullName>Date_validation</fullName>
    <active>true</active>
    <errorConditionFormula>Expiry_Date__c &lt; TODAY()</errorConditionFormula>
    <errorDisplayField>Expiry_Date__c</errorDisplayField>
    <errorMessage>Expiry date cannot be in the past.</errorMessage>
</ValidationRule>
```


Circular Validation


```
<?xml version="1.0" encoding="UTF-8"?>
<CustomObject xmlns="http://soap.sforce.com/2006/04/metadata">
    <actionOverrides>
        <actionName>Accept</actionName>
        <type>Default</type>
    </actionOverrides>
    <actionOverrides>
        <actionName>Accept</actionName>
        <formFactor>Large</formFactor>
        <type>Default</type>
    </actionOverrides>
    <actionOverrides>
        <actionName>Accept</actionName>
        <formFactor>Small</formFactor>
        <type>Default</type>
    </actionOverrides>
    <actionOverrides>
        <actionName>CancelEdit</actionName>
        <type>Default</type>
    </actionOverrides>
    <actionOverrides>
        <actionName>CancelEdit</actionName>
        <formFactor>Large</formFactor>
        <type>Default</type>
    </actionOverrides>
    <actionOverrides>
        <actionName>CancelEdit</actionName>
        <formFactor>Small</formFactor>
        <type>Default</type>
    </actionOverrides>
    <actionOverrides>
        <actionName>Clone</actionName>
        <type>Default</type>
    </actionOverrides>
    <actionOverrides>
```

```
    <actionName>Clone</actionName>
    <formFactor>Large</formFactor>
    <type>Default</type>
</actionOverrides>
<actionOverrides>
    <actionName>Clone</actionName>
    <formFactor>Small</formFactor>
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    <actionName>Delete</actionName>
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    <actionName>Delete</actionName>
    <formFactor>Small</formFactor>
    <type>Default</type>
</actionOverrides>
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    <actionName>Edit</actionName>
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    <enableSharing>true</enableSharing>
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    <externalSharingModel>Read</externalSharingModel>
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    <nameField>
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        <label>Circular Number</label>
  <type>AutoNumber</type>
    </nameField>
    <pluralLabel>Circulars</pluralLabel>
    <searchLayouts></searchLayouts>
    <sharingModel>ReadWrite</sharingModel>
    <visibility>Public</visibility>
</CustomObject>
```


Student Acknowledgement


```
<?xml version="1.0" encoding="UTF-8"?>
<ListView xmlns="http://soap.sforce.com/2006/04/metadata">
    <fullName>All</fullName>
    <filterScope>Everything</filterScope>
    <label>All</label>
</ListView>
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Frontend File


```
<?xml version="1.0" encoding="UTF-8"?>
<Network xmlns="http://soap.sforce.com/2006/04/metadata">
    <allowInternalUserLogin>true</allowInternalUserLogin>
    <allowMembersToFlag>false</allowMembersToFlag>
    <changePasswordTemplate>unfiled$public/CommunityChangePasswordEmailTempl
ate</changePasswordTemplate>
    <communityRoles/>
    <disableReputationRecordConversations>true</disableReputationRecordConversatio
ns>
    <emailSenderAddress>umanoharan44@gmail.com</emailSenderAddress>
    <emailSenderName>CAMPUS CIRCULAR MANAGEMENT</emailSenderName>
    <embeddedLoginEnabled>false</embeddedLoginEnabled>
    <enableApexCDNCaching>true</enableApexCDNCaching>
    <enableCustomVFErrorPageOverrides>false</enableCustomVFErrorPageOverrides>
    <enableDirectMessages>true</enableDirectMessages>
    <enableExpFriendlyUrlsAsDefault>false</enableExpFriendlyUrlsAsDefault>
```

<enableExperienceBundleBasedSnaOverrideEnabled>true</enableExperienceBundle
BasedSnaOverrideEnabled>
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<enableGuestFileAccess>false</enableGuestFileAccess>
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<sendWelcomeEmail>true</sendWelcomeEmail>
<site>CAMPUS_CIRCULAR_MANAGEMENT</site>
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<status>UnderConstruction</status>
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<defaultTab>home</defaultTab>
<standardTab>Chatter</standardTab>
</tabs>
<urlPathPrefix>Circular</urlPathPrefix>
<welcomeTemplate>unfiled$public/CommunityWelcomeEmailTemplate</welcomeTemplate>
</Network>

B. SCREENSHOTS

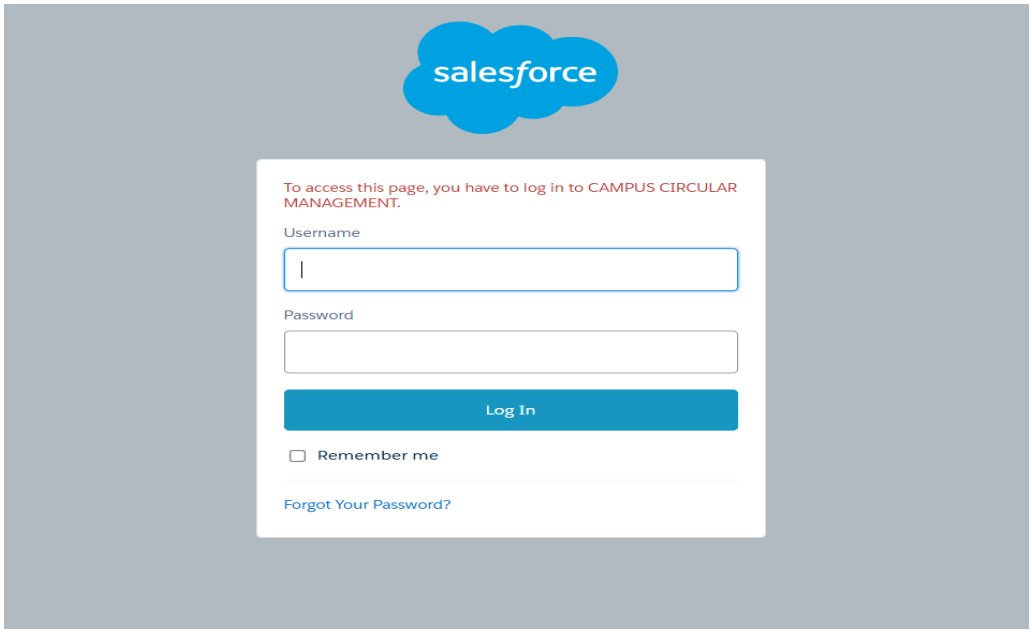

Figure B.1 Login Page

This figure B.1 shows the login page where users enter credentials to securely access the Campus Circular Management System.

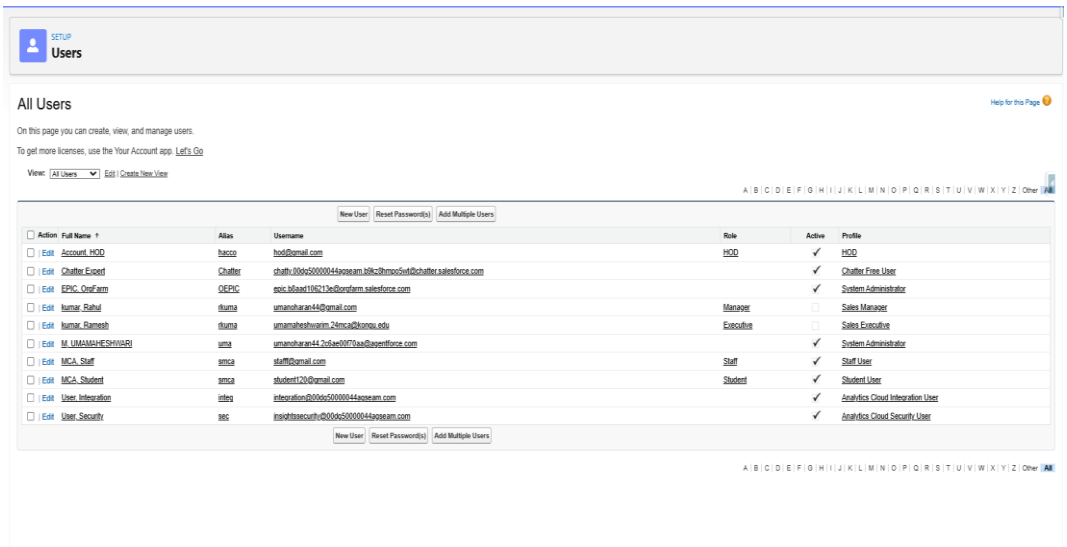

Figure B.2 User Page

This figure B.2 shows the Users Management Page where administrators can view, manage, and control user accounts within the Campus Circular Management System.

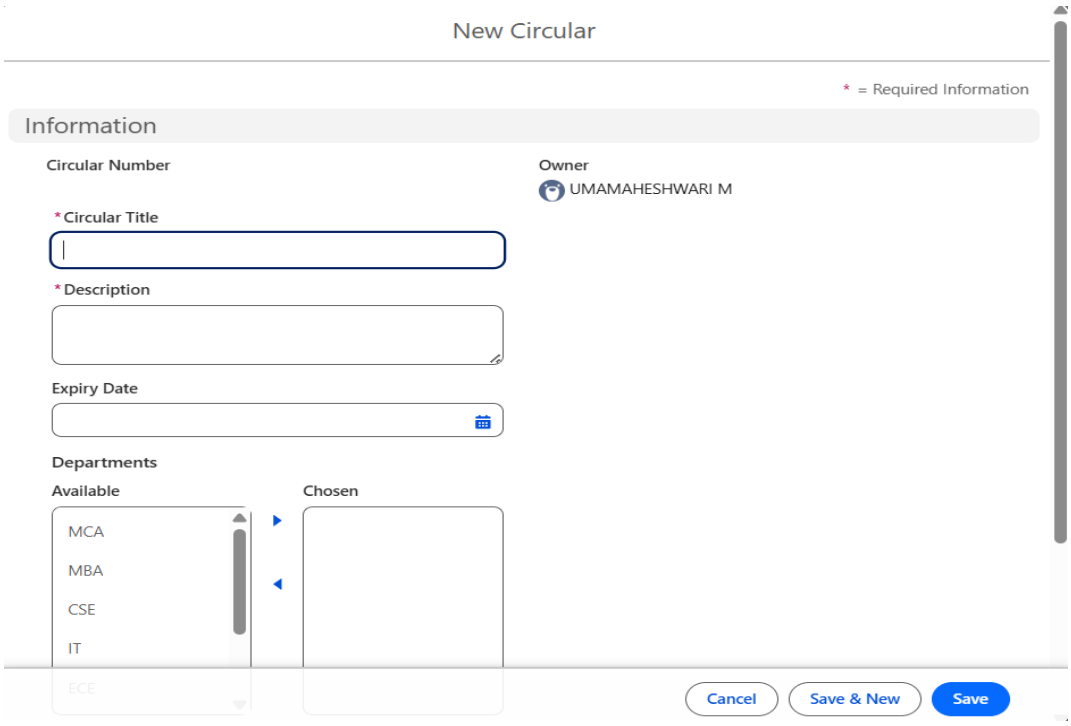

Figure B.3 Circular Page

This figure B.3 shows the Circular Creation Form where the administrator enters details such as title, description, expiry date, and departments to create a new circular.

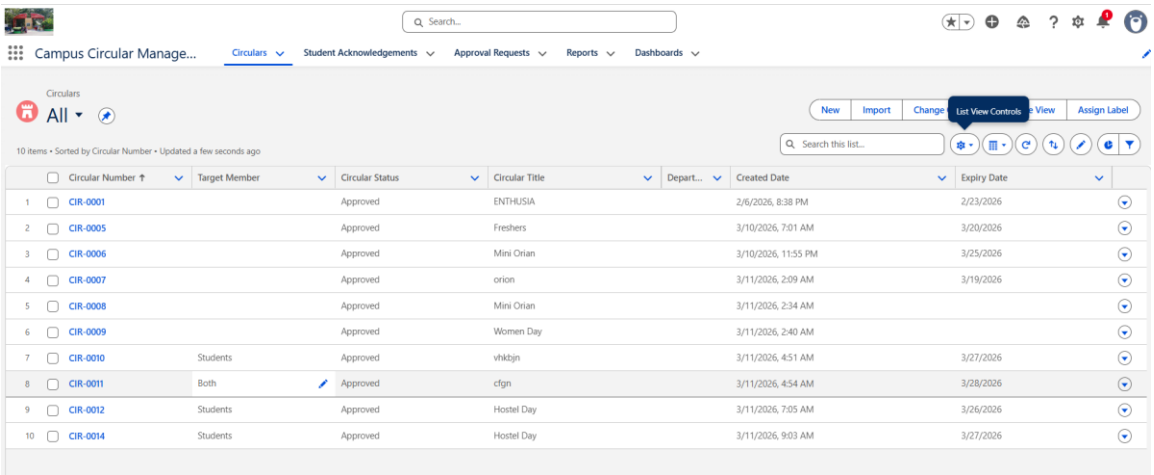

Figure B.4 List of Circular Page

This figure B.4 shows the Circular List Page where all circulars are displayed with details such as circular number, target member, status, and dates for easy management.

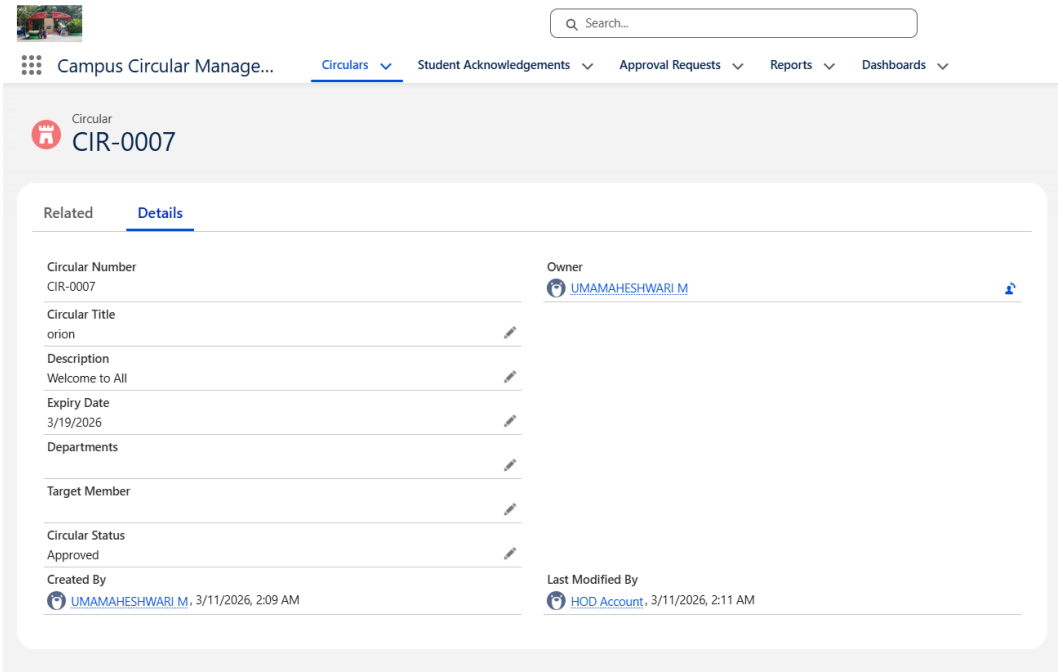

Figure B.5 Circular Detail Page

This figure B.5 shows the Circular Detail Page where complete information about a specific circular is displayed, including title, description, status, and related details.

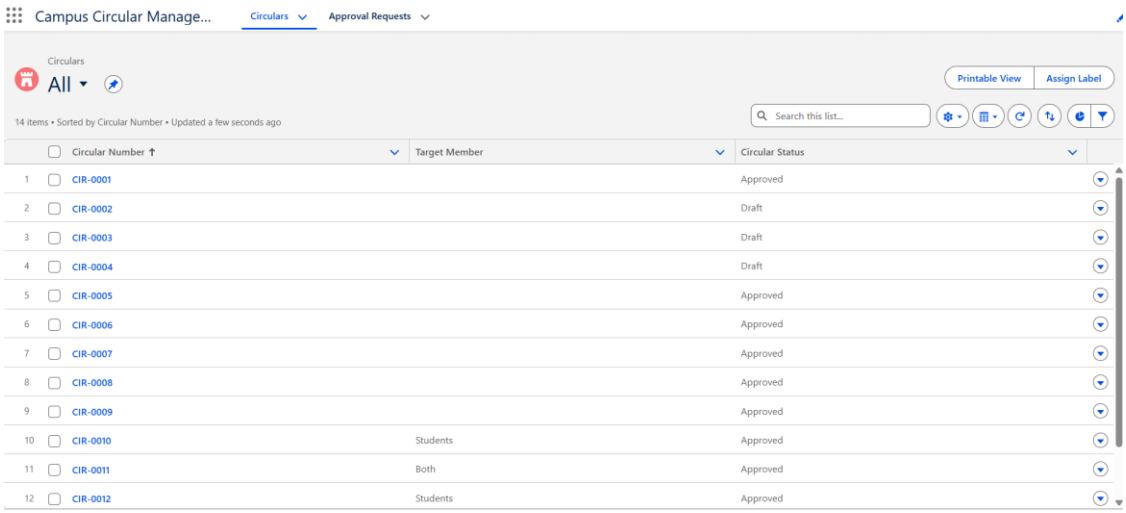

Figure B.6 Approval Page Of Hod

This figure B.6 shows the Circular Approval List Page where circulars are displayed with status such as draft or approved for easy monitoring and approval management

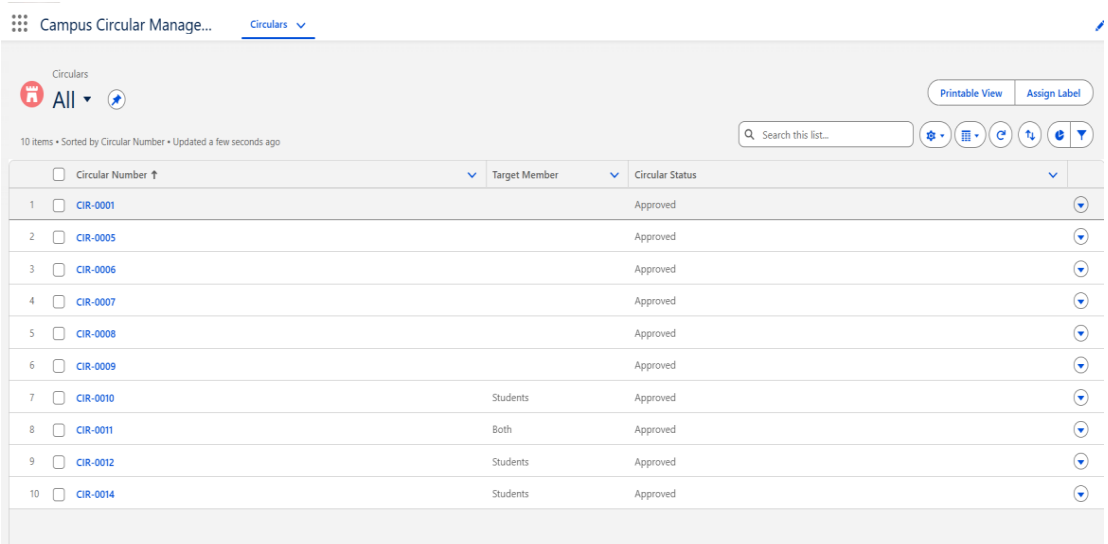

Figure B.7 Student Circular Page

This figure B.7 shows the Approved Circular List Page where all approved circulars are displayed for users to view and manage efficiently.

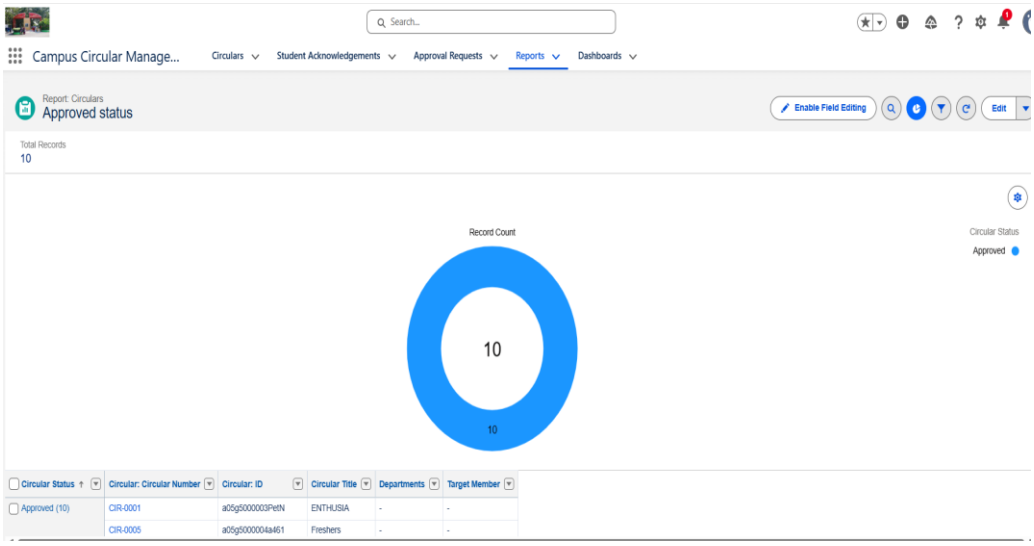

Figure B.8 Approved Circular Record Page

This figure B.8 shows the Approved Circular Report displaying the total number of approved circulars with graphical representation for analysis.

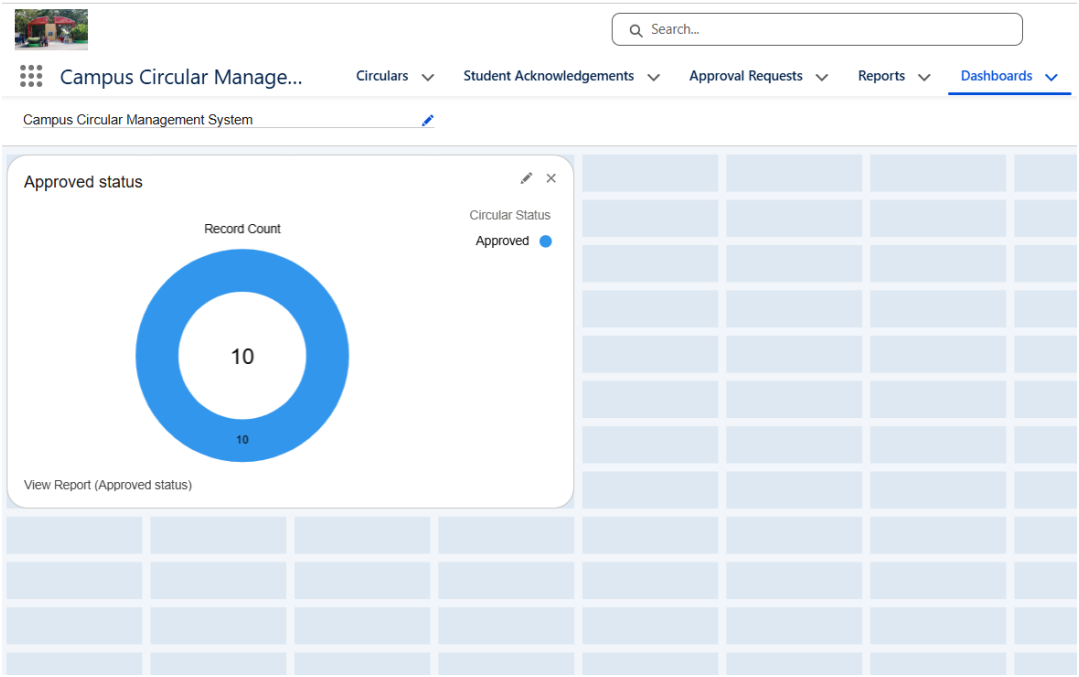

Figure B.9 Dashboard Page

This figure B.9 shows the Dashboard View displaying graphical insights of circular status, helping administrators monitor system activity easily.

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SUSTAINABLE DEVELOPMENT GOALS (SDGS)

SDG 4 – Quality Education

• The Department Circular Management System helps institutions communicate important academic and administrative information efficiently to students and staff.

• Circulars related to examinations, academic schedules, workshops, and seminars can be distributed quickly through the system.

• The platform ensures that students receive timely updates, supporting better participation in educational activities.

• This contributes to improved academic communication and access to educational information.

SDG 9 – Industry, Innovation, and Infrastructure

• The system is developed using Salesforce, providing a modern cloud-based infrastructure for managing institutional communication.

• Digital circular management replaces manual paper-based processes, improving efficiency and reliability.

• Reports and dashboards provide analytical insights that support data-driven decision-making in institutions.

• The project promotes digital transformation within educational organizations.

SDG 16 – Peace, Justice, and Strong Institutions

• The system improves transparency and accountability in institutional communication.

• Circular approvals by the Head of Department (HOD) ensure proper authorization before announcements are distributed.

• Acknowledgement tracking allows administrators to verify that students and staff have received and viewed circulars.

• This structured communication process strengthens institutional governance and information management.